# Introduction to World Models

07/07/2026
Agents reading group
Edwin Brown

# Questions

What are they?

How are they different to video generation?

What problems are they trying to solve?

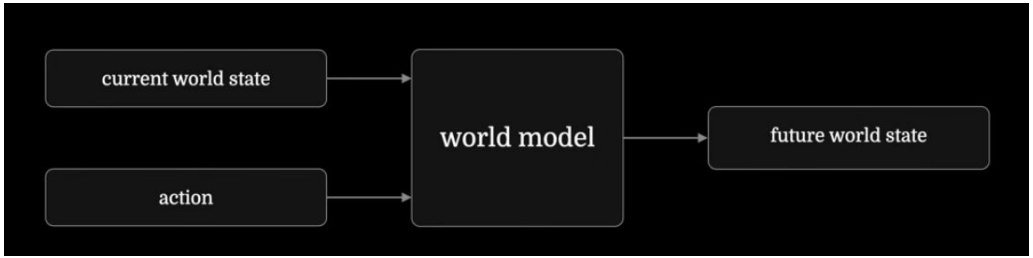
What can currently be done?

Who are the main players?

# What is a world model?

https://worldmodels.github.io/

**Models that make predictions about the physical world**




arXiv:1803.10122v4 [cs.LG] 9 May 2018

## World Models

David Ha [1] Jürgen Schmidhuber [2 3]

**Abstract**

We explore building generative neural network models of popular reinforcement learning environments. Our *world model* can be trained quickly in an unsupervised manner to learn a compressed spatial and temporal representation of the environment. By using features extracted from the world model as inputs to an agent, we can train a very compact and simple policy that can solve the required task. We can even train our agent entirely inside of its own hallucinated dream generated by its world model, and transfer this policy back into the actual environment.

An interactive version of this paper is available at https://worldmodels.github.io

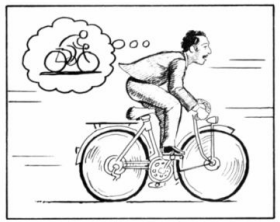


*Figure 1.* A World Model, from Scott McCloud's *Understanding Comics.* (McCloud, 1993; E, 2012)

### 1. Introduction

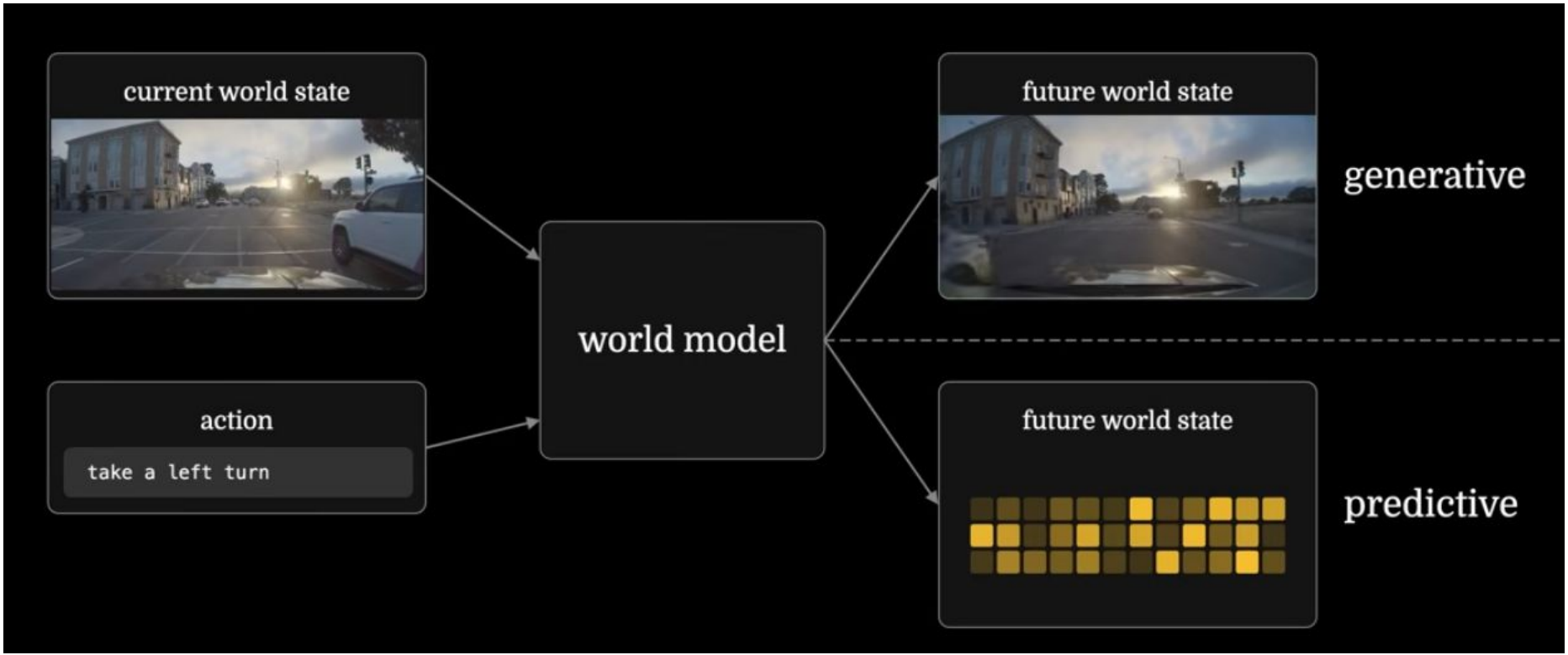
Humans develop a mental model of the world based on what they are able to perceive with their limited senses. The decisions and actions we make are based on this internal model. Jay Wright Forrester, the father of system dynamics, described a mental model as:

*The image of the world around us, which we carry in our head, is just a model. Nobody in his head imagines all the world, government or country. He has only selected*

current motor actions (Keller et al., 2012; Leinweber et al., 2017). We are able to instinctively act on this predictive model and perform fast reflexive behaviours when we face danger (Mobbs et al., 2015), without the need to consciously plan out a course of action.

Take baseball for example. A batter has milliseconds to decide how they should swing the bat – shorter than the time it takes for visual signals to reach our brain. The reason we are able to hit a 100 mph fastball is due to our ability to instinctively predict when and where the ball will go. For professional players, this all happens subconsciously. Their muscles reflexively swing the bat at the right time and loca-

# The definition is full of ambiguity



# Generative vs Predictive

**Generative**

Nvidia - Cosmos Predict

World Labs - Marble

Deepmind - Genie3

**Predictive**

Yann LeCun - JEPA

# Generative - Fei-Fei Li



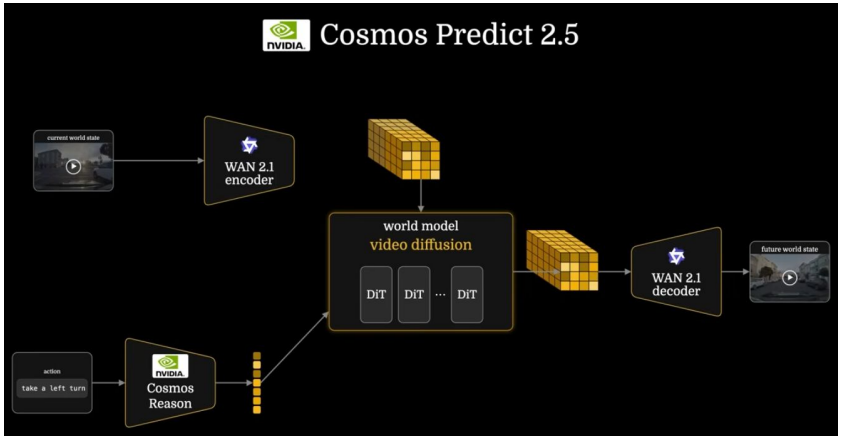
[World Labs]

Main product is [Marble].

Creates explorable persistent 3D spaces

Cannot interact with the environment but can change it in a consistent manner

Gaussian Splatting and Vision Language Action (VLA)



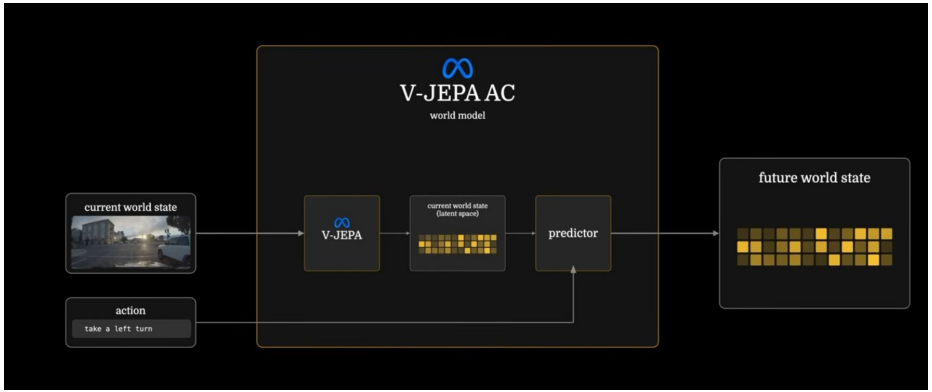
# Predictive - Yann LeCun

[AMI Labs]

Joint Embedding Predictive Architecture (JEPA)

V-JEPA - Video Encoder model - Takes input as input and outputs a latent space.

V-JEPA AC (Action Conditioned) - Uses V-JEPA






A Path Towards Autonomous Machine Intelligence
Version 0.9.2, 2022-06-27

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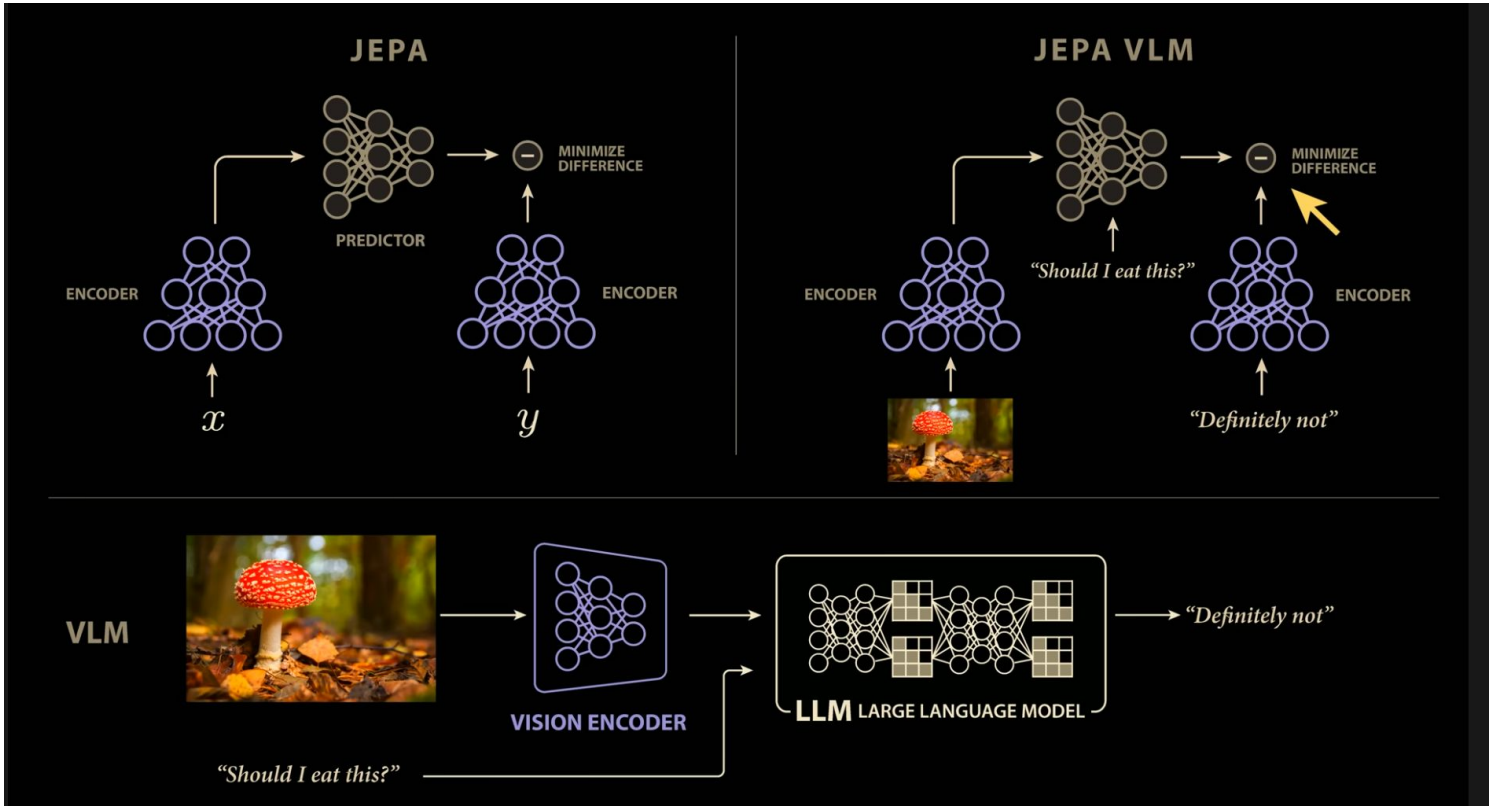
June 27, 2022

**Abstract**

How could machines learn as efficiently as humans and animals? How could machines learn to reason and plan? How could machines learn representations of percepts and action plans at multiple levels of abstraction, enabling them to reason, predict, and plan at multiple time horizons? This position paper proposes an architecture and training paradigms with which to construct autonomous intelligent agents. It combines concepts such as configurable predictive world model, behavior driven through intrinsic motivation, and hierarchical joint embedding architectures trained with self-supervised learning.

**Keywords:** Artificial Intelligence, Machine Common Sense, Cognitive Architecture, Deep Learning, Self-Supervised Learning, Energy-Based Model, World Models, Joint Embedding Architecture, Intrinsic Motivation.

# Vision Language Model vs JEPA

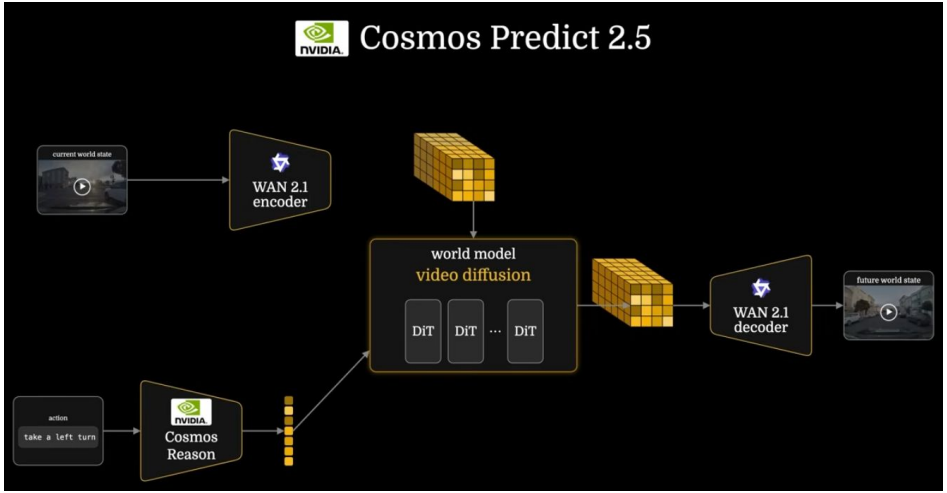


# Generative vs Predictive

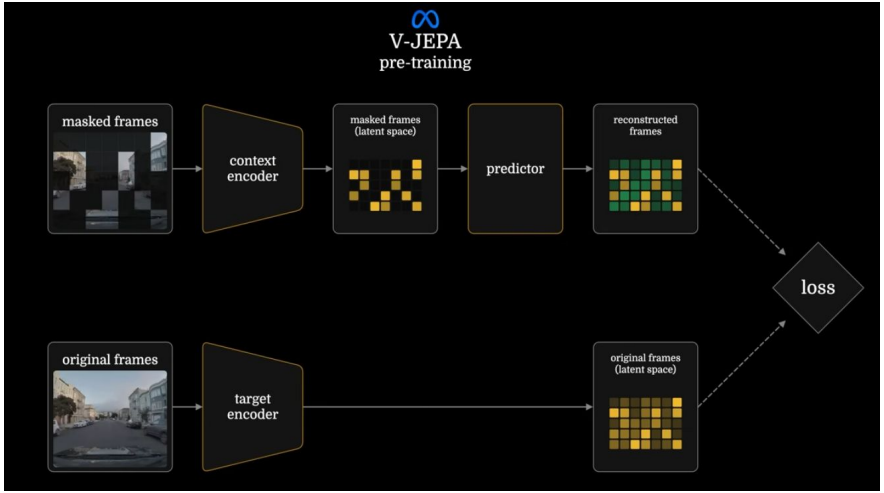
Generative derives its loss in **pixel space.** Predictive derives its loss in **latent space**.

In theory, the encoder only focuses on relevant information and can dismiss inconsequential detail (in pixel space)

Generative : VLA




Predictive : V-JEPA



# Applications

# Synthetic Data Generation

https://wayve.ai/thinking/gaia-2/ - Augmented existing data

https://research.nvidia.com/labs/cosmos-lab/cosmos-predict2.5/ - Multiview generation

https://waymo.com/blog/2026/02/the-waymo-world-model-a-new-frontier-for-autonomous-driving-simulation/ - Not just video but Lidar too!

Probably the most popular use case.

# Interactive Environments

Instead of generating a video, can I navigate inside the world?

Potentially applicable to gaming and entertainment industry.

https://deepmind.google/blog/genie-3-a-new-frontier-for-world-models/ - Behind a pay wall of Genie Max (200 quid a month). Also, 60 second limit, probably because it everything needs to be stored in memory.

“Genie 3’s consistency is an emergent capability. Other methods such as NeRFs and Gaussian Splatting also allow consistent navigable 3D environments, but depend on the provision of an explicit 3D representation. By contrast, worlds generated by Genie 3 are far more dynamic and rich because they’re created frame by frame based on the world description and actions by the user. “

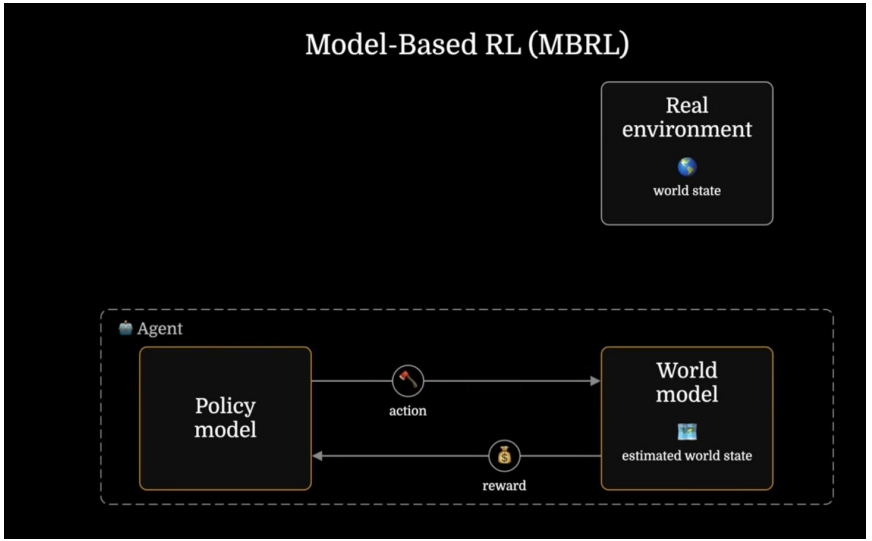
# Training Autonomous Agents

Model-based Reinforcement Learning

As a lot of real environments cannot practically be trained in (autonomous driving for example), world models can provide the training environment.

Playing Doom without prior access to the game
https://worldmodels.github.io/

Robots doing actions without any prior physical testing https://world-gymnast.github.io/




But what if we don’t throw away the world model at inference time!?

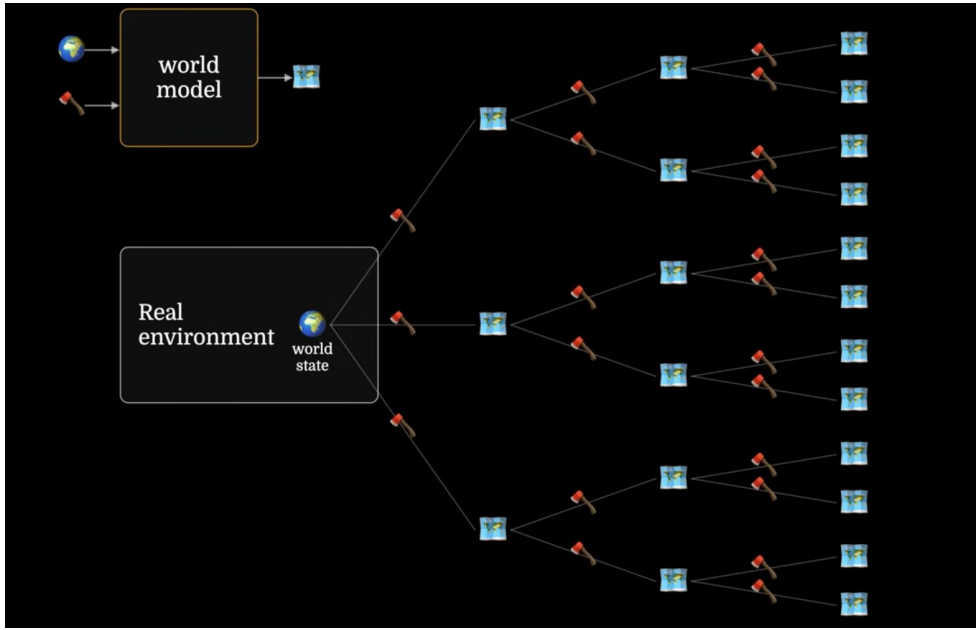
# Inferring with Autonomous Agents

Model Predictive Control

Map out future scenarios and pick the best route

Deepmind MuZero - Go, Atari

Technically, this is what VC JEPA is doing, maybe. .



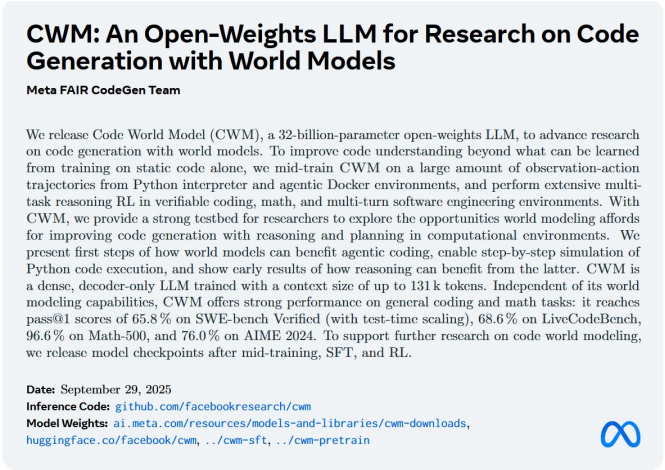

https://arxiv.org/pdf/2506.09985

# Final Thoughts

**“How can we build autonomous systems that cannot predict the consequences of its actions?” -** Yann LeCun

Current Agentic systems basically require a ‘harness’ to deal with the consequences of its action. There is no prediction. Is this a problem?

In theory, this could be done in software domains too, but there less incentive because stuff can be determined deterministically (just run some tests).



**CWM: An Open-Weights LLM for Research on Code Generation with World Models**

**Meta FAIR CodeGen Team**

We release Code World Model (CWM), a 32-billion-parameter open-weights LLM, to advance research on code generation with world models. To improve code understanding beyond what can be learned from training on static code alone, we mid-train CWM on a large amount of observation-action trajectories from Python interpreter and agentic Docker environments, and perform extensive multi-task reasoning RL in verifiable coding, math, and multi-turn software engineering environments. With CWM, we provide a strong testbed for researchers to explore the opportunities world modeling affords for improving code generation with reasoning and planning in computational environments. We present first steps of how world models can benefit agentic coding, enable step-by-step simulation of Python code execution, and show early results of how reasoning can benefit from the latter. CWM is a dense, decoder-only LLM trained with a context size of up to 131 k tokens. Independent of its world modeling capabilities, CWM offers strong performance on general coding and math tasks: it reaches pass@1 scores of 65.8 % on SWE-bench Verified (with test-time scaling), 68.6 % on LiveCodeBench, 96.6 % on Math-500, and 76.0 % on AIME 2024. To support further research on code world modeling, we release model checkpoints after mid-training, SFT, and RL.

**Date:** September 29, 2025
**Inference Code:** github.com/facebookresearch/cwm
**Model Weights:** ai.meta.com/resources/models-and-libraries/cwm-downloads, huggingface.co/facebook/cwm, ../cwm-sft, ../cwm-pretrain

# Resources

https://portfolio.chrisdburr.com/blog/world-models-need-scientists

https://www.youtube.com/watch?v=MqjvfJTCuqw